Your name:

Your student #:

Laboratory day (circle one): Monday Tuesday Wednesday Thursday Friday

Laboratory time (circle one): 9-11am 11am-1pm 2-4pm

TA name (circle one): Gabriel Katie Andrea Bryn Nicola (Niki) Diego

BIOL 300 2024W1 MIDTERM EXAM

For all statistical tests, make sure that you **clearly state each of your hypotheses**. **Unless otherwise stated, assume α = 0.05**. **Show your work** as you solve each problem. For problems that involve multiple steps, the best practice is to avoid rounding between steps. **Be as precise as possible about P-values**. For short answer questions that ask for a numeric answer please **box your final answer**.

By taking this test and writing your name above, you are declaring that your answers on this test are all your own work.

Make sure that your copy of the test includes 7 pages, including this one.

DO NOT MARK THIS TABLE. Exam questions begin on the next page

Pg.	*Points*	*Totals*
2	Q1a ____ Q1b ____ Q1c ____ Q1d ____ Q2 ____ Q3 ____	_____/ 25
3	Q4 ____ Q5 ____ Q6 ____ Q7 ____	_____/ 17
4	Q8a ____ Q8b ____	_____/ 24
5	Q9 ____	_____/ 12
6	Q10 ____ Q11 ____	_____/ 16
7	Q12 ____	_____/ 6
	Grand total:	_____/ 100

1. The numbers below are a sample drawn from a population. All have the same unit of measurement. Express each answer to **two decimal places**.

   23, 48, 19, 12, 36, 91

   a. (*4 points*) What is the sample mean?
   mean = (23 + 48 + 19 + 12 + 36 + 91) / 6 = 229/6
   = 38.1667, therefore rounded answer is **38.17**

   b. (6 *points*) What is the sample standard deviation?
   Sum of squared deviations:
   = (23 – 38.16667)^2 + (48 – 38.16667)^2 + (19 – 38.16667)^2 + (12 – 38.16667)^2 + (36 – 38.16667)^2 + (91 – 38.16667)^2
   = 230.0278 + 96.6944 + 367.3611 + 684.6944 + 4.6944 + 2791.3611
   = 4174.83333

   Divide by n−1 (5 in this case): = 834.9667
   Positive square root = 28.8957, therefore rounded answer is: **28.90**
   Suggested range: 28.88 – 28.92

   c. (*3 points*) What is the median?
   Sorted sample: 12, 19, 23, 36, 48, 91
   Median is the mean of 23 and 36 = (23 + 36) / 2
   Median is **29.50**

   d. (*4 points*) What is the standard error of the mean?
   SEM = s/sqrt(n) = 28.8958/sqrt(6)
   = 28.8958/2.4495 = 11.7967 (if internal rounding is minimal), therefore rounded answer is: **11.80** Suggested range: 11.70 – 11.90

2. (5 *points*) Please describe what a Type I error is. What is the relationship between the Type I error rate and the significance level (α) of a hypothesis test?
   A Type I error occurs when we incorrectly reject the null hypothesis when it is actually true. It is a false positive error. We conclude that there is a statistically significant effect or difference when, in reality, none exists. The significance level (α) of a hypothesis test is the threshold we set to control the probability of making a Type I error.

3. (*3 points*) For a particular study, the Type II error rate is 23%. What is the power of the study?
   Power = 1 – Beta = 77%

Your name: Your student #:

4. (*5 points*) What is a 95% confidence interval for a mean? Please provide as complete a definition as you can.

   A range of values produced by a method such that 95% of confidence intervals produced by the same method contain the population mean. It quantifies uncertainty about the value of the mean based on the data we have collected. It is a range of values surrounding the sample mean that are likely to contain the true mean.
   What would be wrong to say: there is a 95% probability that the population mean lies within the interval.

5. (*4 points*) Select the best definition of "sampling error" from the choices below
   a. Errors that result from having a poor experimental design that introduces a systematic oversampling of a particular segment of a population
   b. The standard deviation of the sampling distribution divided by the population variance
   c. The population variance divided by the cube root of the standard deviation of the sampling distribution
   d. The chance difference between the true population parameter and a sample estimate due to sampling
   e. Errors that result from using a biased sample

6. (*4 points*) Suppose an insurance examiner compared the results of 16462 randomly selected health insurance claims against actual patient records and made a contingency table:

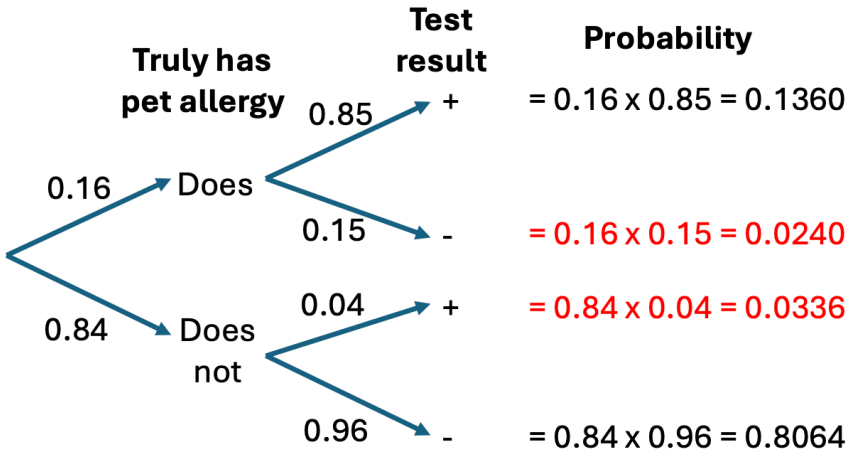
	Accurate claim	Inaccurate but no adjustment	Incorrect, re-coding needed	*Row totals*
Public	8209	2651	4235	15095
Private	146	32	93	271
University	628	190	278	1096
Column totals	8983	2873	4606	16462

   If the examiner were to perform a chi-square contingency test to determine if hospital type (rows) and insurance claim categorization (columns) are independent, how many degrees of freedom would the test statistic have?
   df = (#rows – 1)(#cols – 1)
   = (2)(2) = 4

7. (*4 points*) The law of large numbers states that
   a. As sample size increases, sample standard deviation increases with the square of sample size.
   b. Large numbers contain more degrees of freedom than small numbers
   c. The larger the sample size, the closer an estimate is likely to be to the true parameter value
   d. As sample sizes increase, the distribution of sample means will increasingly approximate a normal distribution
   e. As sample sizes increase, the distribution of sample means will increasingly approximate a normal distribution, but only if the population itself is normally distributed

8. Suppose 16% of children have pet allergies. There is a new at-home test kit for pet allergies on the market that provides families with a convenient way to test if children have a pet allergy (as opposed to some other kind of allergy). The test kit is designed such that a positive result indicates a pet allergy. The probability that the test gives a positive result when a child really does not have a pet allergy is 0.04. The probability that the test gives a negative result when a child really does have a pet allergy is 0.15.

a. (*12 points*) What is the total probability of an incorrect test result? Be sure to think of all the possible ways a test result may be incorrect. Express your answer to three decimal places.



Pr[incorrect test result]=
= Pr[(pet allergy and false negative) or (no pet allergy and false positive)]
=(Pr[pet allergy] Pr[false negative])+(Pr[no pet allergy] Pr[false positive])
=(Pr[pet allergy] Pr[negative result | pet allergy]) +(Pr[no pet allergy] Pr[positive result | no pet allergy])
=0.16×0.15 +0.84×0.04
=0.0576, therefore rounded answer is: **0.058**

b. (*12 points*) Your nephew uses a test kit and sees that he has a positive result. What is the probability that he truly has pet allergies? Express your answer to three decimal places.

Pr[pet allergy | positive result]=
= Pr[positive | pet allergy)*Pr[pet allergy]/ (Pr[positive | pet allergy)*Pr[pet allergy] + Pr[positive | no allergy)*Pr[no allergy])
= 0.85*0.16 / ((0.85*0.16) + (0.04*0.84))
= 0.136/0.1696
= 0.8019, therefore rounded answer is: **0.802**

Your name: Your student #:

9. You are studying for a Chemistry exam and your professor provides an example exam from 2015 with an answer key to help you study. As you are going through it, you get the sense that this professor has an affinity for making certain letters the correct answer to multiple choice questions more often than others. You decide to formally test this. You go through the answer key and determine the frequency of correct letter choices (see table).

Choice	# of times it was the correct choice
A	21
B	35
C	19
D	23
E	22

(*12 points*) Please perform a hypothesis test of whether these letter choices have equal probability of being the correct choice. Use $\alpha = 0.05$ for your significance level. Please state your hypotheses explicitly and show all work. Based on the evidence you have gathered, what would you conclude?

H_0: The frequencies of correct letter choices are proportional to the number of answer choices (p = 0.2)
H_A: The frequencies of correct letter choices are not proportional to the number of answer choices (p not equal to 0.2)

120 total questions -> Expected = 120/5 = 24

$$\chi^2 = \frac{(21-24)^2}{24} + \frac{(35-24)^2}{24} + \frac{(19-24)^2}{24} + \frac{(23-24)^2}{24} + \frac{(22-24)^2}{24}$$
$$\chi^2 = \frac{9}{24} + \frac{121}{24} + \frac{25}{24} + \frac{1}{24} + \frac{4}{24}$$
$$\chi^2 = \frac{160}{24} = 6.6666$$

df = 5 – 1 = 4

Critical value from table at alpha = 0.05 -> 9.49

With an observed χ^2 of 6.666 at alpha 0.05 and 4 df, we fail to reject the null hypothesis that the frequencies of correct answers are proportional to the number of answer choices

10. (*4 points*) Suppose you are working with a data set of wingspans for 510 individuals of *Haliaeetus leucocephalus* (bald eagle). The standard deviation of these wingspans is 0.082 meters, the sample mean is 2.050 meters, and the data follow a normal distribution. Please rank the following confidence intervals of the mean wingspan in order of *narrowest to widest interval range*.

    95% confidence interval; 50% confidence interval; 99% confidence interval

    a. 95% CI, 50% CI, 99% CI
    b. 99% CI, 95% CI, 50% CI
    c. 50% CI, 95% CI, 99% CI
    d. This is a trick question. All intervals would have the exact same width.

11. (*12 points*) Based on the data set in Question 10, about what proportion of bald eagles have wingspans between 1.894 and 2.196 meters? Express your answer to three decimal places.

    $Z1 = (1.894-2.050)/0.082 = -1.902$
    $Pr[Z < -1.092] = Pr[Z > 1.902] = 0.02872$

    $Z2 = (2.196-2.050)/0.082 = 1.780$
    $Pr[Z > 1.780] = 0.03754$

    $Pr[-1.092 < Z < 1.780] = 1 - 0.02872 - 0.03754$
    $= 0.93374$, therefore rounded answer is **0.934**

Your name: Your student #:

12. (*6 points*) Is a P-value a conditional probability? Please state yes or no and explain your reasoning by providing a complete definition of P-values
Yes, it is. It is the probability of observing a test statistic as extreme or more extreme than the test statistic you actually observed, given the null hypothesis being true. It is conditional on the null hypothesis being true.